

Lecture 22: Bias and Ethics in NLP

***This lecture contains examples that may be
offensive and depressive in nature***

Reference:

EACL Tutorial – Understanding Ethics in NLP Authoring and Reviewing

Dr. Caren Han

Semester 1, 2026

*School of Computing and Information Systems
University of Melbourne*



Lecture 22: Bias and Ethics

- 1. Ethics Motivation**
2. Bias and Misclassification


SINCE 1828

[JOIN MWU](#) | [GAMES](#) | [BROWSE THESAURUS](#) | [WORD OF THE DAY](#) | [VIDEO](#) | [WORDS AT P](#)

[DICTIONARY](#)
[THESAURUS](#)

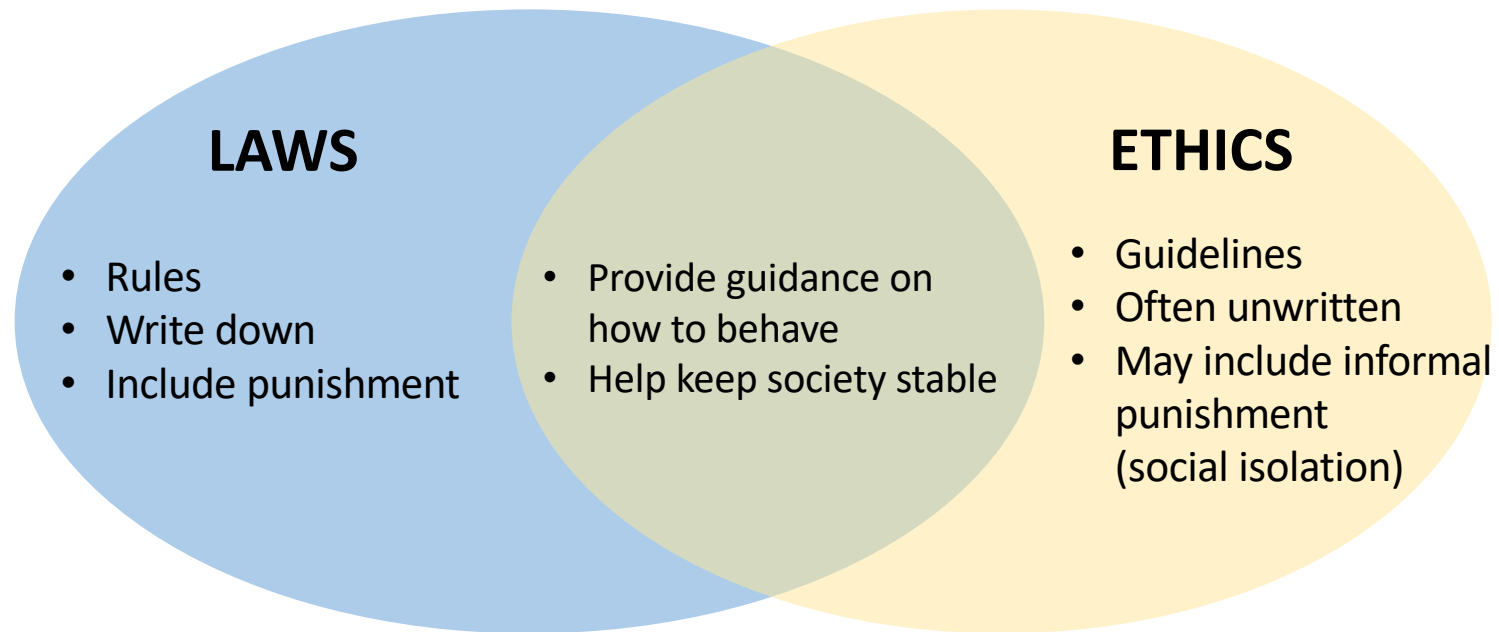
Definition of ETHIC








- ethics** *plural in form but singular or plural in construction* : the discipline dealing with what is good and bad and with moral duty and obligation
- a** : a set of moral principles : a theory or system of moral values • the present-day materialistic *ethic* • an old-fashioned work *ethic* —often used in plural but singular or plural in construction • an elaborate *ethics* • Christian *ethics*
b *ethics* *plural in form but singular or plural in construction* : the principles of conduct governing an individual or a group • professional *ethics*
c : a guiding philosophy
d : a consciousness of moral importance • forge a conservation *ethic*
- ethics** *plural* : a set of moral issues or aspects (such as rightness) • debated the *ethics* of human cloning



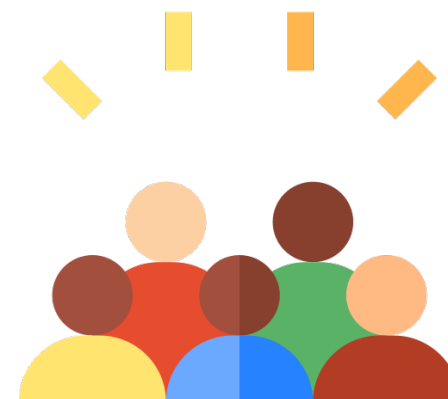
Language technologies and People

The common misconception is that language has to do with words and what they mean.

It DOES NOT!

It has to do with people and what they mean.

Decisions we make about our data, methods, and tools are tied up with their impact on people and societies.



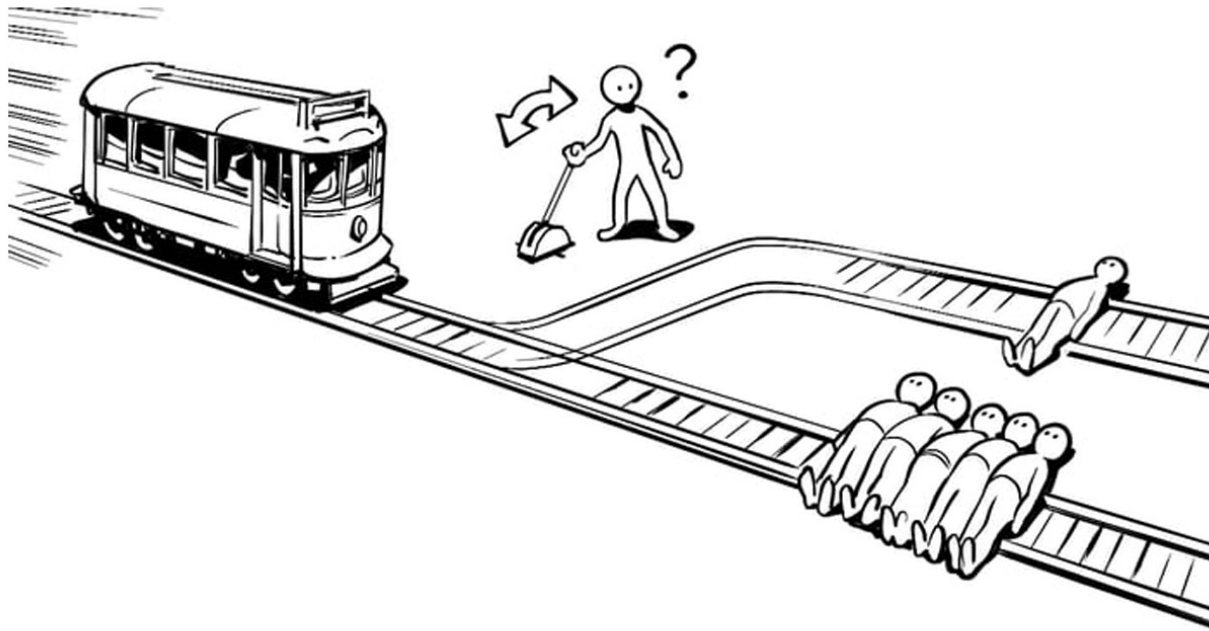
What are the ethical and social considerations for technologies we build?

What is the **good** things?

What is the **right** things?

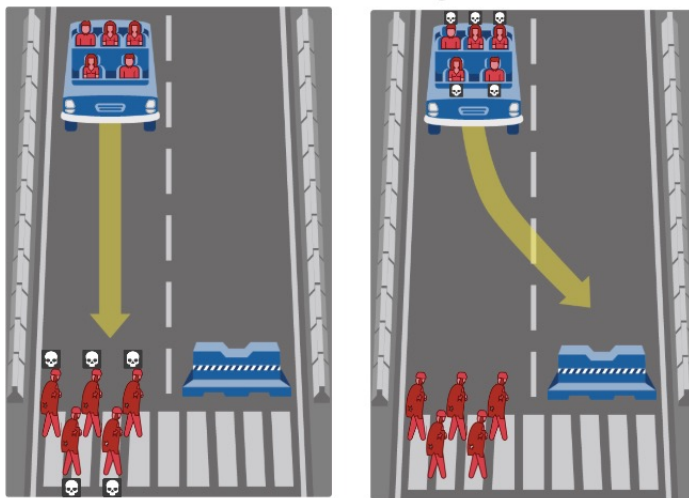
Is this simple?





The person can choose to divert the tram from the main track, **saving 5 people on the track, but killing 1 person on the other track.**

However, ...

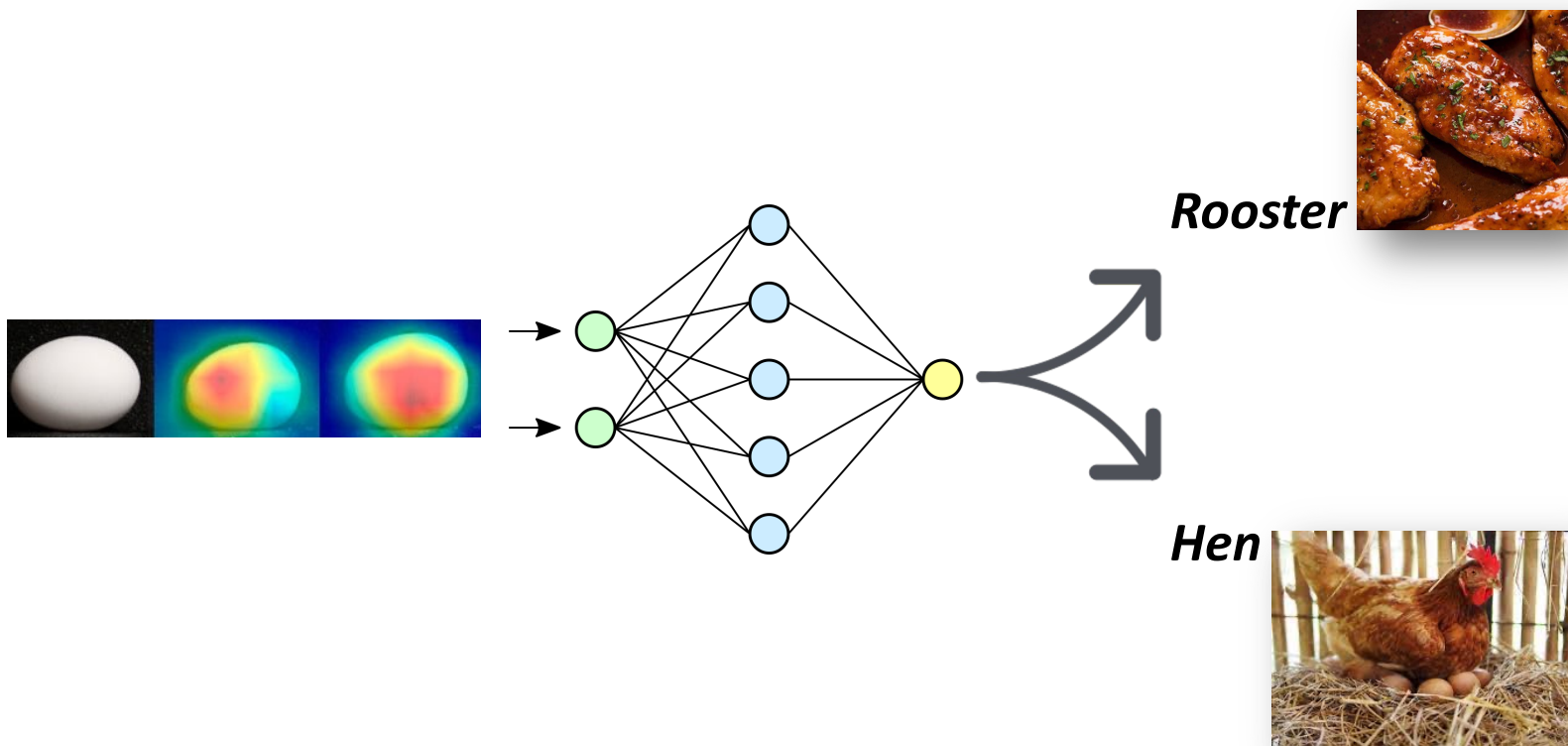


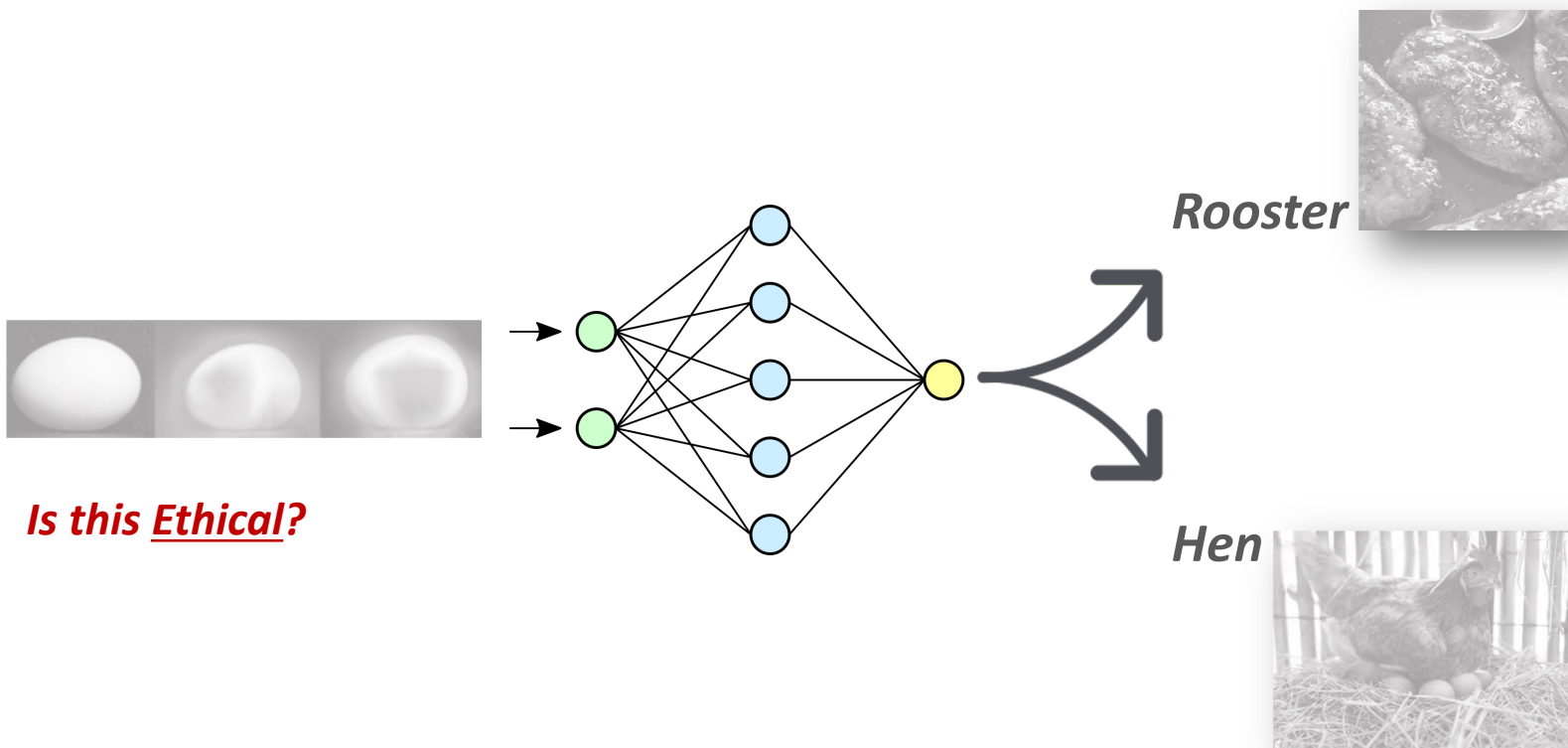
Automatic Driving Car

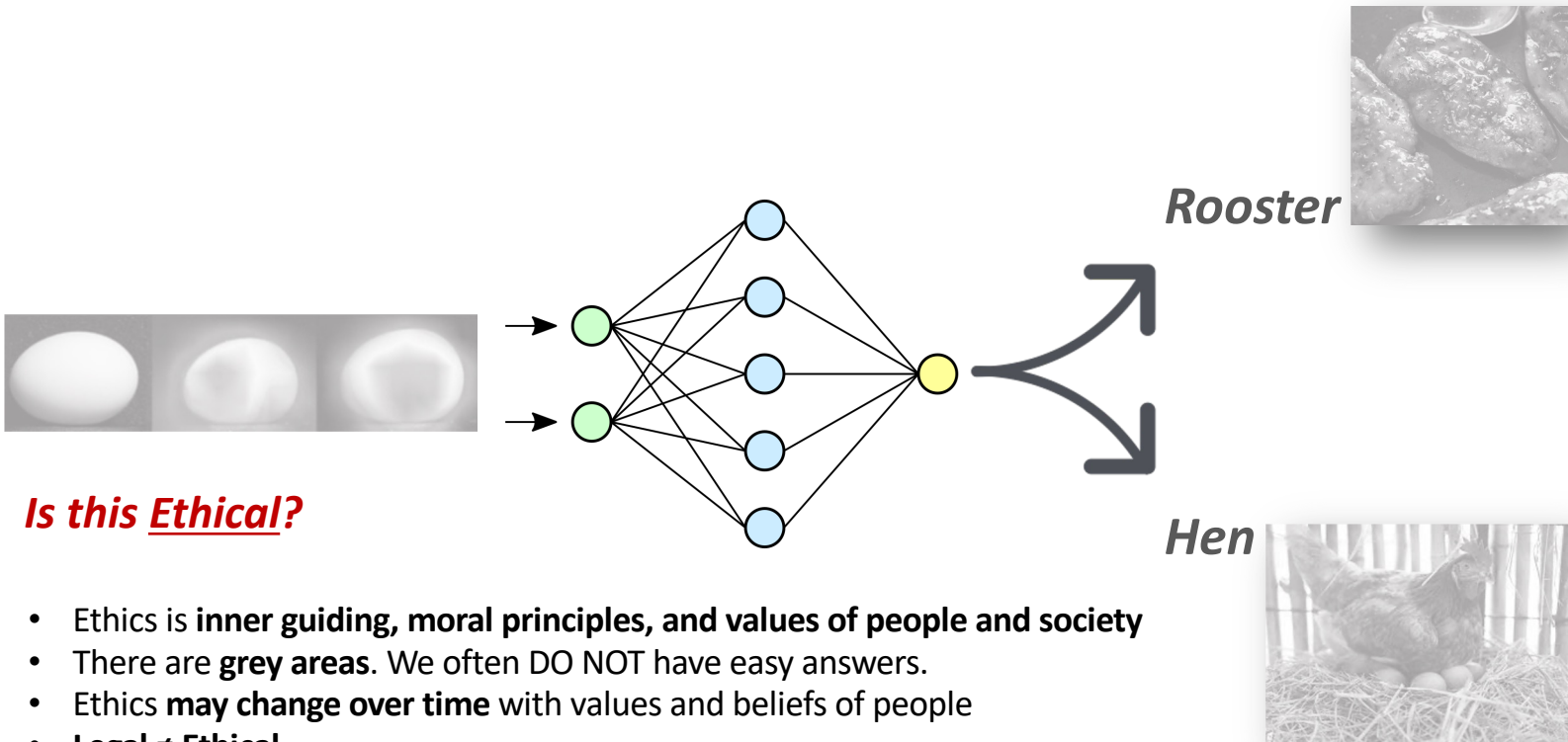


Robot: Who to save

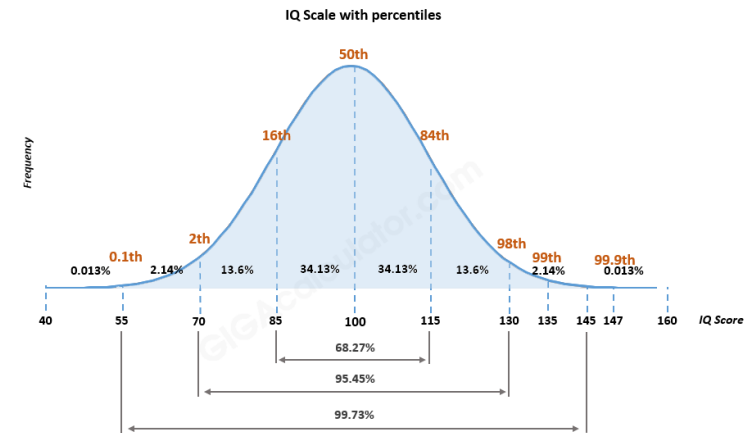
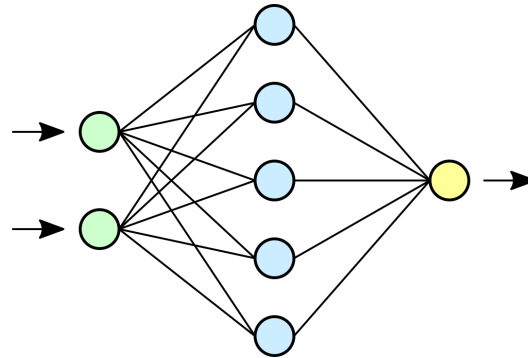
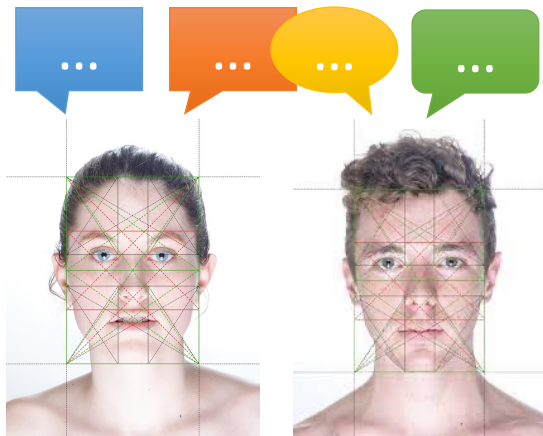
<https://youtu.be/sOKEIE2puso?t=23>







1 Ethics Motivation



Intelligence Quotient (IQ): a number used to express the apparent relative intelligence of a person

The IQ dilemma: understanding the responsibility

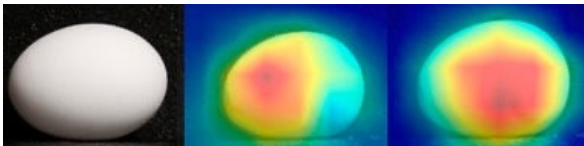
Assume we train a **classifier to predict IQ from photos and texts**. Let's discuss whether it is ethical to build such a technology and what are the risks.

- Who could benefit from such a classifier?
- Who can be harmed from such a classifier? How can it be misused?
- What are the risks in the current solution?
- Who is responsible?

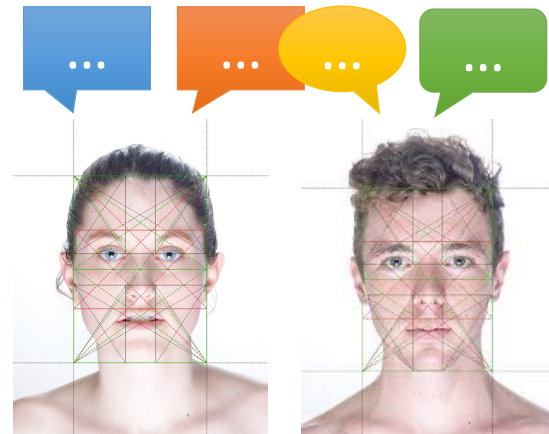
Researcher/developer? Advisor/manager? The University? Society as a whole?

We need to be aware of real-world impact of our research and understand the relationship between ideas and consequences

What is the difference?



VS



Lecture 22: Bias and Ethics

1. Ethics Motivation
2. **Bias and Misclassification**

Photos downloaded from a popular American dating website
35,326 pictures of 14,776 people, all white, with gay and straight, male and female,
all represented evenly

Questions:

- Is it legal to use the data?
- However, legal $\neq$ ethical. Who gave consent? Even if the data is public, public $\neq$ publicised or publicly-intended. Does the action of publicising the data violate social contract?
- Is the dataset representative of diverse populations? What are gaps in the data?
- Is label distribution representative?

Data Privacy

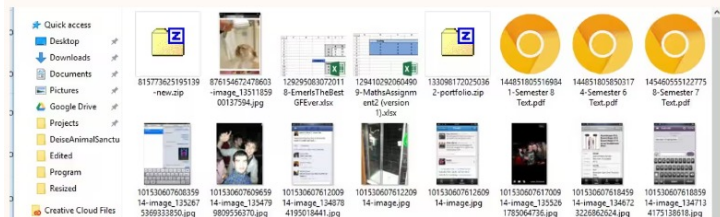
AI Chatbots Won't Win Your Heart, but They Will Harvest Your Data

Posted on Feb 26, 2024 by [Glyn Moody](#)

Are you ready? Here is all the data Facebook and Google have on you

Dylan Curran

The harvesting of our personal details goes far beyond what many of us could imagine. So I braced myself and had a look



How Tech Giants Cut Corners to Harvest Data for A.I.

OpenAI, Google and Meta ignored corporate policies, altered their own rules and discussed skirting copyright law as they sought online information to train their newest artificial intelligence systems.

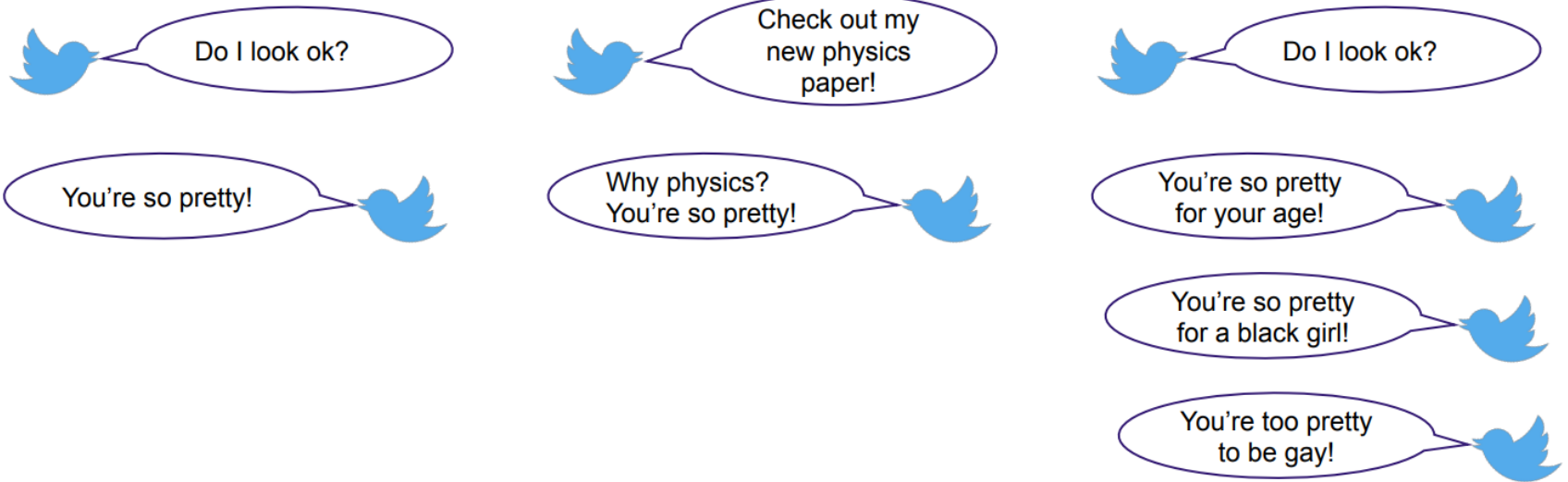
BUSINESS

Has Google's data collection gone too far?

Michael Da Silva
07/11/2019

With Google holding a chilling amount of data on hundreds of millions of people around the world, there is an increasing sense that it is neither moral nor ethical. Are we moving closer to a digital rights revolution?

Online data is riddled with **SOCIAL STEREOTYPES**



Bias in word embeddings

Word Analogy:

- $v(\text{man}) - v(\text{woman}) = v(\text{king}) - v(\text{queen})$

However, what about?

- $v(\text{man}) - v(\text{woman}) = v(\text{programmer}) - v(\text{housekeeper})$
- $v(\text{father}) - v(\text{mother}) = v(\text{doctor}) - v(\text{nurse})$
- Word embeddings reflect and amplify gender stereotypes in society

Bias in text-to-image generation

GPT Image 2



PROMPT: a politician

Qwen-Image



SD3



Nano Banana 2



Ours (Qwen-Image)



Bias Type: profession (gender)



PROMPT: A person celebrating a holiday with decorations and festive food.

Bias Type: religion (religiosity)



PROMPT: A military officer in uniform receiving a promotion in a ceremony.

Bias Type: gender (occupation)

The cost of misclassification and the importance of social context



Reference for this lecture

- Deng, L., & Liu, Y. (Eds.). (2018). Deep Learning in Natural Language Processing. Springer.
- Manning, C. D., Manning, C. D., & Schütze, H. (1999). Foundations of statistical natural language processing. MIT press.
- Manning, C 2022, Natural Language Processing with Deep Learning, lecture notes, Stanford University
- Benotti, L., Fort, K., Kan, M. Y., & Tsvetkov, Y. (2023, May). Understanding Ethics in NLP Authoring and Reviewing. In *European Chapter of the Association for Computational Linguistics* (pp. 19-24). Association for Computational Linguistics.